

Terms & The Terminal

Ctrl+Alt+T - Shortcut to open the terminal

- Terminal - The CLI environment, also called a shell
- Directory - a folder in Windows
- Execute - run a command
- “~” - A “tilde”, stands for your home directory.
 - This can be used instead of typing out the full path in the terminal
- Path - specific location of a file/directory, uses \ in Windows but a / in Linux
 - Example: `/home/jrodriguez/Downloads/file.txt`

Basic Commands

- **cd**
 - Change directory
 - Example: `cd ~`
 - This command will move you to the home directory (the main directory for your user)
- **touch**
 - Creates a file
 - Example: `touch ~/Downloads/file.txt`
- **ls -al**
 - `ls` - list, will print the contents of a directory
 - - `al` - this option will print permissions, size, and other info about files/directories
 - Example: `ls -al ~/Downloads`
 - Will list the files in your Downloads directory
- **cp**
 - Copy
 - Example: `cp ~/Downloads/photo.jpg ~/Pictures/`
 - This command will copy “photo.jpg” *from* the Downloads directory *to* the Pictures directory
- **mv**
 - Move
 - Example: `mv ~/Downloads/photo.jpg ~/Pictures/`
 - This command will move the photo from the Downloads directory to the Pictures directory

- **rm**
 - o Remove
 - o Example: **rm ~/Downloads/photo.jpg**
 - Deletes the photo
 - Files deleted this way are not recoverable!
 - **Never execute this command:**
 - **rm -rf /**
 - o **THIS WILL DELETE THE ENTIRE SYSTEM**
- **| (pipe)**
 - o This is the symbol you get from Shift+\ on the keyboard
 - o This redirects the output of a command to another command
- **grep**
 - o Grep lets you search for a specific “string”
 - o Example: **ls ~/Downloads | grep “photo.jpg”**
 - Let me break this down
 - **ls ~/Downloads** will list the contents of the Downloads folder
 - **|** will pass the list printed by the **ls** command to the next command
 - **grep** will search for “photo.jpg” and print what it finds.
- **cat**
 - o This command will print the contents of a file
 - o Example: **cat ~/Downloads/file.txt**
- **nano**
 - o Nano is the text editor installed by default on Ubuntu. There’s also Vi and Vim, but they’re complicated.
 - o Example: **nano ~/Downloads/file.txt**
 - This opens the file and lets you edit it, sort of like Notepad in a terminal
- **sudo**
 - o Runs a command AS ADMINISTRATOR
 - Example: **sudo apt update**
 - This command will update your system.
 - You will have to enter your password to confirm sudo commands (every few uses, or in a new terminal)

Options & --help

To get info on any command/tool you can use the **-h** option. If that fails you can try **- --help**. These are called *options*. I think they call them flags in Windows, and they look like this \? with a slash instead of a hyphen/dash.

Ng1, these help pages can be kind of hard to read sometimes. So just know that whenever you see code like this: **Command <options> [file]** you don't need to include the <>[] symbols. You can also view the "manual pages" for a command like this:

man command

Options change how a command will execute. Here's an example:

cp -r

The **-r** option makes the command copy "recursively". This lets you copy a directory *and* its files instead of just the directory.

Networking Commands

- **ip link**
 - o Gives you information about network interfaces
 - Most importantly if they're up or down
- **ip address**
 - o Gives you the same information, plus IP address for interfaces
- **ip route**
 - o Displays the routing information for the device
 - Most importantly default gateway, subnet, and subnet mask for interfaces
- **ping**
 - o Pretty sure you know this one
 - o Useful for testing connectivity and TTL (Time-to-Live)
- **traceroute**
 - o Useful for seeing which hops traffic follows
 - o Will print the IP (if possible) of routers the packet passes through on the way to it's destination
 - o Can help find a point of failure in a network

Updating & Installing Software

- **sudo apt update**
 - o Will update the package list on your system
- **sudo apt upgrade**
 - o Will actually update the system
- **sudo apt update && sudo apt upgrade**
 - o Will do the whole update thing
- **sudo apt install program**
 - o Will install the Program named "program"

Operators

Operators are symbols that changes what happens after a command is executed. Sort of. Basically. I already dropped a couple of these, so hopefully this'll make some sense.

- | (pipe)
 - o Example: `cat ~/Downloads/file.txt | grep "phrase"`
 - This use cat to print the file, then grep searches the output for the string "phrase"
- &&
 - o This lets you execute multiple commands (one after the other)
 - o Example: `touch file.txt && cp file.txt ~/Downloads/`
 - This will create the file and then copy it to the Downloads folder
- >
 - o This is a "redirect"
 - o This will send the output of a command somewhere else
 - o Example: `ls Downloads > downloads.txt`
 - Lists the files in Downloads and sends that to downloads.txt
 - Will create a file, or overwrite what's already in a file
- >>
 - o "Append" the output of a command to a file
 - o Adds it to the end of the file
 - o Won't overwrite a file
 - o Example `ls Documents >> downloads.txt`
 - Will list the files in Documents and add it to the end of downloads.txt